MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

MINOR PROJECT SYNOPSIS

Multi-Modal ML Framework for Car Resale Price Prediction

Submitted by

Name	Registration No	Department
Arnav Kiran Shetty	230905318	SCE
Arthik Shetty	230905249	SCE
Ansh Raj	230905492	SCE

Under the Guidance of:

Dr. KAMARAJ K

School Of Computer Engineering

Manipal Institute of Technology, Manipal, Karnataka – 576104

Minor Specialization: Computational Intelligence

Offered by: School of Computer Engineering

1. Introduction

In the rapidly evolving automotive landscape, the secondary market for used vehicles has seen an unprecedented surge, driven by increasing consumer aspirations and the rising costs of new vehicles. While online platforms have streamlined the process of finding used cars, determining an accurate and fair resale value remains a significant challenge. Currently, most valuation models operate on a "one-size-fits-all" basis, relying solely on primary attributes like the manufacturing year, brand popularity, and odometer readings. This conventional approach, however, often overlooks the most critical factor: the actual internal health and maintenance history of the specific vehicle.

The primary hurdle in the current used-car market is **information asymmetry**. A vehicle that appears aesthetically maintained may possess an engine near mechanical failure, while another with higher mileage might be in superior condition due to meticulous servicing. Standard machine learning models treat these vehicles as identical, leading to financial risks for buyers and unfair pricing for sellers. Furthermore, the wealth of qualitative data stored using sensors and service logs is often written in unstructured formats or technical jargon, hence is typically ignored by traditional algorithms.

This project introduces an intelligent **Multi-Modal ML Framework** designed to move beyond basic statistical averages. By augmenting standard market features with **mechanical health indicators** and **Natural Language Processing (NLP)** of service records, the system aims to quantify the "hidden" condition of a vehicle. Through the analysis of maintenance patterns and estimated **Remaining Useful Life (RUL)** metrics, the framework provides a more transparent, condition-adjusted valuation. This holistic approach ensures that the predicted resale price reflects not just the car's chronological age, but its actual mechanical integrity and the quality of its upkeep over time.

2. Literature Review

The challenge of accurate vehicle valuation has transitioned from simple statistical models to integrated systems that consider both market trends and technical integrity. A recent study by **Bergmann and Feuerriegel (2025)** emphasizes the importance of granular vehicle equipment such as navigation systems and alloy rims finding that including such high-dimensional data improves price prediction performance by 3.27%[2]. Their research utilizes an automated procedure to map over 50,000 equipment identifiers into 18 categories, highlighting that specific configurations are key determinants of market value.

However, a significant gap remains in capturing the dynamic mechanical health of a vehicle. While traditional models by **Lessmann and Voß (2017)** and **Gleue et al. (2019)** establish baseline valuations using mileage and age, they often treat "overall condition" as a coarse variable[5,6]. This project addresses this by integrating **Prognostic Health Management (PHM)**.

A critical component of this framework is the use of the **NASA C-MAPSS dataset**, originally intended for turbofan engines. Although designed for aviation, researchers have proven its validity for automotive contexts. **Chao et al. (2021)** demonstrate that the physics of degradation, specifically the thermal and vibration "wear-out" pattern follow universal laws of thermodynamics and friction that are shared between jet turbines and internal combustion engines[3,4]. By utilizing the C-MAPSS dataset, the model can learn to recognize these universal signatures of mechanical decay, which can then be mapped to car engine health through **Cross-Domain Transfer Learning**.

Furthermore, while the ETH Zurich study uses NLP to identify static equipment , the proposed system utilizes DistilBERT to extract "maintenance sentiment" from unstructured service logs[7]. This bridges the gap between the mechanical "truth" found in sensor data and the qualitative history recorded by technicians, providing a truly multi-modal valuation.

3. Problem Statement

Existing car valuation methods are largely superficial and reactive in nature. They rely predominantly on historical market averages, brand popularity, and odometer readings to estimate price. However, these factors fail to account for the internal mechanical state of the vehicle. By the time a buyer discovers a significant engine issue or maintenance neglect, the transaction has often already been completed, leading to "The Market for Lemons" effect, where information asymmetry results in financial loss for the buyer or undervaluation for a well-maintained seller.

Moreover, current systems primarily focus on quantitative data (age and mileage) without considering the qualitative evidence found in service records or the physical health of the vehicle. They fail to capture the nuances of a car's history, such as the difference between "highway miles" and "city miles," or the impact of consistent versus sporadic maintenance. Traditional models cannot "read" mechanic notes or "interpret" sensor data, leaving a massive gap in the valuation process.

This project aims to address these limitations by developing a **Multi-Modal ML Framework** that analyzes both market trends (structured data) and mechanical integrity (unstructured data). By combining **Natural Language Processing (NLP)** for service record analysis with **Prognostic Health Indicators** derived from engine metrics, the system seeks to proactively adjust the resale value based on the car's actual condition. This ensures a more accurate, transparent valuation that reflects the true remaining life and quality of the vehicle before the point of sale.

4. Objectives

i. To develop a robust valuation model that utilizes historical resale datasets to establish a baseline market price for used vehicles.

ii. To analyze textual maintenance records using NLP techniques, specifically identifying patterns such as frequent repairs, high-cost component replacements, and the overall quality of service history.

iii. To extract and analyze diagnostic features from engine data, such as vibration patterns and thermal trends, to identify mechanical wear or potential failure points.

iv. To implement a condition-based scoring system that quantifies the vehicle’s mechanical health and Remaining Useful Life (RUL) into a standardized index.

v. To combine market trends with mechanical analysis (multi-modal approach) to improve the accuracy and reliability of price prediction compared to systems relying solely on mileage and age.

vi. To provide a transparent valuation report for users by generating a condition-adjusted price and a health summary, helping prevent unfair pricing and information asymmetry in the used-car market.

5. Methodology

The project follows a multi-branch architecture designed to fuse economic market data with real-time technical diagnostics:

Step 1: Data Collection The system utilizes diverse data sources for a holistic valuation:

- Market Data: Historical used-car listings (e.g., Kaggle/Scraped data) containing price, mileage, and brand attributes.
- Textual Data: Synthetic and scraped service records, mechanic notes, and maintenance logs.
- Mechanical Data: Real-time diagnostic data via **OBD-II (On-Board Diagnostics)** protocols for current health snapshots, supplemented by the **NASA C-MAPSS dataset** for modeling long-term engine degradation.

Step 2: Data Preprocessing

- Tabular market data is cleaned, handled for missing values, and categorical variables are encoded.
- Textual logs undergo tokenization and normalization to handle technical automotive jargon.
- OBD-II and Sensor signals are standardized through resampling, normalization, and noise reduction to ensure consistency across different vehicle models.

Step 3: Feature Extraction

- Mechanical Features**:** We extract **Time-Domain** (Mean, Standard Deviation) and **Frequency-Domain** (via Fast Fourier Transform) features from the C-MAPSS trajectories to identify "spectral signatures" of engine decay.
- Textual Features**:** Contextual embeddings are generated using **DistilBERT [2]**, converting qualitative notes into a "Maintenance Sentiment Vector."
- Market Features**:** Standard categorical encoding is applied to car brands and fuel types.

Step 4: Model Development

- A Gradient Boosting Regressor (XGBoost) is trained on market data to predict the baseline Fair Market Value (FMV).
- A Deep Learning model (LSTM or 1D-CNN) is trained on sensor trajectories to predict the Remaining Useful Life (RUL) and quantify mechanical wear.
- An NLP classifier is fine-tuned to convert maintenance logs into a numerical "Condition Score."

Step 5: Multi-Modal Fusion

The outputs from the Market, NLP, and Mechanical branches are integrated using a Late Fusion architecture. This approach is inspired by multi-modal frameworks, where the baseline price is dynamically adjusted by a multiplier derived from the fused diagnostic metrics.

Step 6: Report & Value Prediction

The framework generates a comprehensive valuation report using all 3 modes of data provided. Adjusting the market value with the mechanical health of the vehicle.

6. References

[1] A. Saxena and K. Goebel, "Turbofan Engine Degradation Simulation Data Set," *NASA Ames Prognostics Data Repository*, 2008.

[2] S. Bergmann and S. Feuerriegel, "Machine learning for predicting used car resale prices using granular vehicle equipment information," *Expert Systems with Applications*, vol. 263, 125640, 2025.

[3] M Arias Chao, C Kulkarni, K Goebel, O Fink "Aircraft Engine Run-to-Failure Dataset under Real Flight Conditions for Prognostics and Diagnostics"

[4] S. Lessmann and S. Voß, "Car resale price forecasting: The impact of regression method, private information, and heterogeneity on forecast accuracy," *International Journal of Forecasting*, vol. 33, no. 4, pp. 864-877, 2017.

[5] V. Sanh et al., "DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter," *arXiv preprint arXiv:1910.01108*, 2019.